Support Updating and Candidate Re-decision Across Seven Medical Question-Answering Systems

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Abstract

Medical question answering requires deciding when changed information adds support, invalidates a premise, alters candidate rankings, or leaves a judgment intact. We evaluate seven system configurations spanning six baseline families on a frozen benchmark of 13,905 inputs per configuration and 6,104 non-reference native units from 11 sources. Error analysis motivates two reusable terminal components: question-scoped rule execution and original-model re-decision within a public diagnostic catalog. These components improve every configuration on the complete R1, R2, and R3 development strata, containing 2,866, 79, and 733 units. Gains range from 14.83–19.12, 16.46–29.11, and 18.69–35.20 percentage points, respectively. The strata overlap; they are not independent replications. Matched controls show substantial benefits from the catalog interface, variable additional effects of voting and calibration, and a prescribing head that preserves some remaining support but trails a same-coverage constant-answer control in aggregate accuracy. We then study the CF-MoA composition and additional candidate verification in a six-arm experiment on the two MedRAG configurations. Its adopted core improves the full native-unit mean by 10.09 and 12.01 points. Rationale-conditioned verification changes that mean by -1.81 and $+0.93$ points, with 95% paired family intervals of $[-2.53, -1.08]$ and $[+0.30, +1.56]$. The study establishes consistent terminal-component gains across seven development configurations while separating those gains from the narrower, mixed effect of additional verification. Small external component evaluations delimit transfer; independent confirmation of the final composition remains unexecuted.

Keywords: medical question answering; counterfactual evaluation; support updating; candidate verification; retrieval-augmented generation

1 Introduction

A medical question may change without requiring its answer to change. An added observation can strengthen one diagnosis while leaving another diagnosis preferable; removing one contraindication can leave an independent contraindication intact. Conversely, a seemingly small change to a patient attribute or a stated evidence condition can require a different answer. These cases place different demands on a question-answering system. They also make simple measures of answer agreement inadequate: identical answers can reflect either appropriate invariance or a failure to update.

Retrieval-augmented generation (RAG) supplies language models with external information (Lewis et al., 2020), and medical RAG examines how retrieval choices interact with question-answering performance (Xiong et al., 2024). Retrieved knowledge still has to be applied to the correct patient, option, and question. Our starting point is therefore a comparison of seven systems: direct answering, MedRGAG, MA-RAG, MedRAG with Llama and with Qwen,

MedGENIE, and TC-RAG. These are seven configurations of six baseline families, using two underlying language-model families. All seven cover the same 13,905-input v5 benchmark under its frozen native evaluation contract.

The comparison reveals low scores on newly established support (R1), withdrawn support (R2), and candidate reconsideration (R3), but their errors have different structures. Prescribing examples include a numerical threshold applied incorrectly despite the relevant rule being present, and facts from a retrieved patient's history assigned to the current patient. Diagnostic errors combine incorrect disease decisions with names, lists, or diagnostic granularity that the frozen scoring vocabulary does not map. These findings motivate separate, testable interventions: extract current-patient facts and execute question-scoped rules, or let the original model make a new decision in an explicit public candidate space.

We implement these interventions as reusable terminal heads, preserving the original model weights and permitted inputs. The diagnostic head operates on each system's actual saved context; the rule head combines facts generated by the corresponding backbone with that system's initial answer. All seven configurations improve on each of the complete R1, R2, and R3 development strata. These are substantial system-level results: R1 gains range from 14.83 to 19.12 percentage points, and R3 gains from 18.69 to 35.20 points. Component controls clarify their interpretation. Single-pass catalog selection explains much of the diagnostic gain, while additional voting and prior correction have mixed effects. The prescribing component improves over every original system and retains some still-supported options, although a same-coverage constant "none" policy achieves higher aggregate accuracy on this imbalanced stratum.

We next examine how these established components behave in a broader composition. CF-MoA routes visible inputs to support recomputation, diagnostic comparison, a conditional model executor, or repeated native-answer readout. Its five role names describe operations; support addition and withdrawal share one executor. This later stage uses the two MedRAG configurations, with six arms per configuration on full v5. It does not replace the seven-system experiment or constitute a seven-system evaluation of the complete CF-MoA pipeline. Sampling and aggregation methods (Wang et al., 2023, 2025) motivate the further question of whether an extra candidate-selection stage improves an already developed core.

The resulting evidence has three levels. First, we provide a common native-task comparison of seven systems and trace their different failure modes. Second, we show consistent development gains from terminal support recomputation and catalog re-decision across those systems, with coverage, explicit component controls, and bounded external component checks. Third, the two-configuration CF-MoA study separates composition gains from additional verification: the core improves the full native-unit mean by 10.09 and 12.01 points, whereas rationale-conditioned verification changes it by -1.81 and $+0.93$ points. Complete source, paired-correctness, failure, and cost analyses support these distinctions. The v5 results are exposed-development evidence; broader clinical utility and independent confirmation of the final composition remain separate empirical questions.

2 Related Work

Retrieval and medical question answering. RAG combines generation with access to retrieved documents (Lewis et al., 2020). MedRAG and the Medical Information Retrieval-Augmented Generation Evaluation (MIRAGE) benchmark examine retrieval-augmented medical question answering across corpora, retrievers, and language models (Xiong et al., 2024). Our baseline set additionally includes local adaptations of MedRGAG (Li et al., 2026), MA-RAG (Wu et al., 2026), MedGENIE (Frisoni et al., 2024), and TC-RAG (Jiang et al., 2025), alongside direct answering. These systems represent different retrieval, generated-context, and multi-step decision procedures. Their original corpora, computation budgets, and model interfaces differ;

the common benchmark compares concrete system configurations rather than isolating one retrieval algorithm. Our terminal interventions retain the allowed context of each configuration and evaluate how available information is converted into a native answer. MedRAG appears twice because both Llama and Qwen configurations are included, and these two configurations also support the later CF-MoA composition study.

Repeated reasoning and aggregation. Self-consistency aggregates sampled reasoning paths (Wang et al., 2023). Mixture-of-agents uses layered agent outputs as information for subsequent generation (Wang et al., 2025). CF-MoA's adopted core instead selects an applicable operation and preserves its native result; its name identifies the local system lineage rather than an architectural equivalence to layered mixtures. The candidate-verification extension permits a bounded comparison of answers already generated by one component. An expanded-pool voting arm and a joint-selection arm distinguish this extension from simply generating more answers or selecting from the same pool in one decision.

Candidate presentation and calibration. Contextual calibration studies the influence of content-free inputs on language-model predictions (Zhao et al., 2021). The diagnostic component uses a fixed disease catalog, several candidate permutations, and an existing content-free prior correction. That component belongs to B. The additional verifier's two code mappings serve a different purpose: they average binary judgments under opposite correctness encodings. These scores remain uncalibrated ranking signals. Neither procedure turns a model score into a probability of clinical correctness.

Medical perturbation and evidence benchmarks. The evaluation combines existing medical diagnosis, prescribing, question perturbation, evidence substitution, and robustness sources. Their distinctions matter: a changed patient fact, an irrelevant contextual cue, and a task that explicitly stipulates counterfactual evidence impose different answer contracts. Section 3 and Appendix B identify the source artifacts. We use these artifacts as the basis of a frozen local evaluation and retain their source-specific scoring limitations. The literature selection is targeted background for this empirical study, not a systematic review or a claim that adjacent approaches are absent.

3 Task and Evaluation Contract

3.1 Inputs and decision operations

An inference input is $x = (q, o, e, c, a_0)$, where q contains the current question and patient information, o contains native options when present, e is the required supplied evidence, c is the original system's allowed context, and a_0 is its saved answer when available. An explicit executable model is an additional optional input. Each arm receives the same legal input for a given model configuration. Gold answers, evaluation labels, classification explanations, hidden paired endpoints, and source identifiers that could provide an answer shortcut are excluded from inference. Provenance and scoring information remain in the offline evaluation ledger.

The benchmark uses five overlapping labels attached to specific judgments. R1 denotes newly established support; R2 denotes withdrawn or contradicted support; R3 denotes a need to compare competing judgments again; R4 denotes a consequence that must be inferred from changed premises; and R5 denotes a judgment that should remain unaffected. Support changes do not imply that the final answer must flip. R1–R3 may coexist, and R5 is established by the task's meaning rather than by model agreement. The labels describe evaluation strata; routing never reads them.

The system operates on each visible endpoint separately. It is not given both members of a hidden evaluation pair and asked to recover the expected relation. Pair-level measurements are constructed only after native predictions have been scored.

The project protocol permits external MedCorp retrieval while retaining all required source evidence. In particular, MedCounterFact inputs preserve their stipulated evidence, and MedRGB inputs retain their six fixed evidence-ratio conditions while also permitting external retrieval. The latter is a project adaptation of MedRGB, rather than a reproduction of its original static-evidence-only setting. Original and modified endpoints have the same information-access permissions. The present study reuses the saved contexts under that protocol.

3.2 Native units and denominators

The frozen v5 evaluation contains 13,905 unique inference inputs per model configuration, 15,616 native scoring mappings, and 6,104 non-reference evaluation units from 4,766 families. One input can support multiple mappings, and one unit can contain several native tasks or variants. Reference inputs support paired evaluation but are excluded from the main ALL score. These counts are not interchangeable sample sizes, and none is described as a count of independently recruited patients.

Construction retained every eligible R1–R4 target in the preceding classified collection. For R5, 450 eligible comparison units were randomly sampled without replacement from each of eight source strata, using a seed generated once before selection and sorted unit identifiers. Model predictions and prior v3 membership did not determine these draws. The resulting R5 evaluation scope has 3,600 units. Frozen evaluation membership is distinct from the original semantic labels: an unsampled R5 label attached to a retained R1 unit does not automatically add that unit to the R5 score.

Let J_u be the non-reference native scoring records belonging to unit u, and let $z_{muj} \in \{0, 1\}$ be native main-answer correctness for arm m. We use the frozen unit score and overall mean

$$s_{mu} = \frac{1}{|J_u|} \sum_{j \in J_u} z_{muj}, \qquad S_m = \frac{1}{6104} \sum_{u=1}^{6104} s_{mu}. \tag{1}$$

The main metric is $100S_m$. It is not a source macro-average, an average of the five R-category scores, or input-level accuracy. Native rules include exact option keys, exact sets for the primary multi-select score, normalized relation labels, and a frozen canonical diagnostic mapping. Multi-select overlap and robustness auxiliary fields are separate diagnostics. The diagnostic mapping is a local scoring contract, not the original source's semantic language-model grader. For R4 clinical scenarios, correctness means agreement with the source's expected answer.

The R-category denominators are R1: 2,866; R2: 79; R3: 733; R4: 13; and R5: 3,600. Because these groups overlap, their counts and scores cannot be summed to recover ALL. The five input formats contain 3,560 diagnosis inputs, 167 multi-select inputs, 989 relation inputs, 2,700 robustness inputs with auxiliary output, and 6,489 other single-choice inputs.

The earlier terminal-head experiments report the complete R1, R2, and R3 strata. Each unit in these three strata has one non-reference scoring task, so their reported correct counts divided by 2,866, 79, and 733 equal their corresponding category means. All 733 R3 units belong to R1; R1 and R2 intersect in four units, including three R3 units. Thus, 3,678 category memberships represent 2,941 unique units. Seven system evaluations of a shared unit are repeated system observations, not seven independent patients. These category-specific results do not define a seven-system CF-MoA ALL score.

3.3 Paired correctness and uncertainty

The frozen paired ledger contains 6,838 maintain pairs and 1,780 respond pairs. Their expected relations are based on the native task and reference answers, with compatible answer meanings,

rather than inferred mechanically from R labels. For pair p with endpoints x_p^0 and x_p^1, the main paired score is

$$t_{mp} = \mathbb{1}[a_m(x_p^0) = y_p^0]\,\mathbb{1}[a_m(x_p^1) = y_p^1]. \tag{2}$$

Thus, a model receives credit only when both endpoints are correct; repeating an incorrect answer does not count as successful maintenance. Native scoring, rather than literal string equality where inappropriate, determines each indicator.

For each later full-v5 CF-MoA contrast, 5,000 paired bootstrap samples draw families with replacement using seed 240924. All compared arms share a resampled family sequence. Each replicate averages the atoms contributed by the sampled families, preserving unit weights within families instead of assigning equal weight to family means. ALL uses 6,104 units in 4,766 families; maintain uses 6,838 pairs in 2,806 families; respond uses 1,780 pairs in 1,780 families. We report 95% percentile intervals. These intervals describe variation in the exposed benchmark and do not correct for method selection, multiple comparisons, or previous data access.

The seven-system baseline and terminal-head tables report their complete observed point estimates. We do not attach the later CF-MoA bootstrap intervals to those earlier comparisons or treat overlapping category improvements as independent replications.

Families group the related variants of an original case; cross-source groups with exactly matching original questions and options are connected into a dependency group. This procedure does not establish that all semantic near-duplicates have been identified.

4 Support Updating and Candidate Re-decision

4.1 Reusable terminal heads

The first development stage adds terminal decision components to each original system. A head is an external inference procedure, with no newly trained parameters. It receives only the current input, the permitted original context where needed, the original answer, and a callback to the original backbone. Visible task format and rule eligibility select the branch. R labels select evaluation strata offline and never select an answer or a branch at inference. The R1 and R3 interfaces combine the two components below and retain the original answer on uncovered tasks; the R2 interface uses the rule component alone.

Shared support addition and removal. The rule component separates patient-fact extraction from execution of a fixed public rule bank. Its established eligibility predicate is limited to supported multi-select prescribing questions, including an explicit age of at least 65 and a recognized rule-bank drug option. The existing bank contains 71 rules; question scope determines the applicable rules and requested actions. Facts are extracted from the current question, not from unrelated patients in retrieved documents. Missing facts remain unknown.

Each supported option receives a state in $\{\text{MET}, \text{CONTRADICTED}, \text{UNKNOWN}\}$. Necessary conditions combine by conjunction; independent support paths combine by disjunction. The existing documented-exception policy and native assembler remain authoritative. For an ordinary valid initial option set A_0, away from native special cases, the update is

$$A_{\text{new}} = \{o : s(o) = \text{MET}\} \;\cup\; \big(A_0 \cap \{o : s(o) = \text{UNKNOWN}\}\big). \tag{3}$$

The expression adds newly supported options, removes options whose relevant support is contradicted, and preserves selected options with unresolved support. Native "none" exclusivity, invalid initial answers, and extraction failures use the established assembler and fallback policies. The same execution can add newly supported options and withdraw unsupported ones, allowing the frozen component to serve both R1 and R2 phenomena. Positive-field extraction

and evidence sentence identifiers aid traceability; the existence of a cited sentence alone does not establish that it entails the extracted fact.

Diagnostic comparison. The diagnostic component uses a fixed public catalog of 49 diseases derived from the DDXPlus resource (Fansi Tchango et al., 2022). The original model scores candidate codes under six permutations. Existing content-free distributions provide a prior correction for each permutation; the corrected choices are combined by majority vote, with mean corrected probability and fixed catalog order resolving ties. The frozen diagnostic scorer still uses its existing 46-label mapping. We do not expand that mapping after observing predictions. Catalog constraints, candidate resources, repeated decisions, and calibration all contribute to this component, so its overall gain cannot be attributed to calibration alone.

The catalog head preserves the original system's actual final decision context and presents the current case again before scoring the fixed candidates. It does not retrieve training-case labels or replace decisions with a nearest-neighbor answer. The six permutations, prior correction, and tie policy are shared across configurations, including configurations on which one component decreases accuracy. A separate single-permutation control and uncorrected six-vote control measure the contribution beyond catalog re-decision itself.

4.2 CF-MoA adopted core B

The later CF-MoA composition incorporates the established terminal heads and additional operations. It first routes diagnosis questions to the catalog component, then eligible prescribing questions to the rule component. Inputs with a complete supported executable model can use the model-execution component. Supported single-choice and relation questions with their original message context use repeated native-answer readout, denoted F. Other inputs preserve the original answer. In the two-configuration full-v5 evaluation, 3,560 inputs route to diagnosis, 123 to rules, 10,178 to F, and 44 to passthrough per configuration. The explicit-model component covers zero clinical-v5 inputs. A1 and A2 record support addition and withdrawal from the shared rule execution; their names do not introduce two independent fact extractors.

Conditional model execution. The A4 component extracts a typed intervention and executes it only when the complete supported physiological model, parameters, state, and question contract are supplied. Its historical validation uses a fixed glucose simulation model and is reported separately in Appendix D. The clinical v5 data do not meet this capability predicate. The 13 R4-labeled clinical units nevertheless remain in the evaluation and follow the other applicable routes.

Repeated native-answer readout. For ordinary supported tasks, F produces up to five complete native answers using seeds 42–46 and stops once a key has the required absolute majority of three votes. If all five draws complete without that majority, plurality among valid keys selects the answer, with ties resolved by the earliest response; if no key is valid, the first invalid response is retained for failure accounting. Invalid outputs remain in the trace. A separate adopted natural language inference (NLI) path uses two reasoning paths and two code mappings; this path is retained unchanged by the candidate-verification extension. In both cases, the adopted component's complete native response remains the output authority. B has no subsequent global generative editor.

4.3 Rationale-conditioned candidate verification

The extension operates only on the ordinary F branch. Let $\mathcal{C}(x)$ contain the distinct legal answer keys observed in the saved F pool. For each key, the earliest complete saved response is its representative. Invalid votes remain recorded but do not become candidate keys. Candidate membership does not use gold correctness, expected improvement, or offline R labels. F must provide an available legal result and at least two distinct legal candidates before verification

runs.

For candidate k, the verifier retains the original messages, native options, and allowed evidence. It appends the target candidate and the observed candidate set. In the rationale-conditioned arm, it also appends the earliest saved reasoning associated with each candidate, explicitly identified as untrusted model opinion. It does not append vote counts, the F winner as an authoritative label, hidden paired inputs, or evaluation metadata. A rationale is neither a newly retrieved source nor a verified patient observation.

Two scoring calls use opposite binary code mappings. Let $\ell_{kh}(b)$ be the log score of code $b \in \{A, B\}$ for mapping $h \in \{1, 2\}$, and let b_h^+ be the code meaning "the target candidate is correct." The ranking score is

$$v(k \mid x) = \frac{1}{2} \sum_{h=1}^{2} \frac{\exp \ell_{kh}(b_h^+)}{\exp \ell_{kh}(A) + \exp \ell_{kh}(B)}. \tag{4}$$

We select the maximum-scoring candidate. An exact tie retains the F key when it is tied; otherwise native candidate order breaks the tie. The complete representative response is returned without generating a new answer or repairing auxiliary fields. A completed verification makes $2|\mathcal{C}(x)|$ logical binary-scoring requests. Physical missing-code retrieval and failed-call costs are separately recorded and can make physical request counts differ from this expression. The score v is an uncalibrated ranking signal.

Outside the supported branch, with fewer than two legal candidates, or on the adopted NLI route, the system retains B. A candidate-scoring failure records a technical failure and uses the saved F answer; an unavailable F answer remains unavailable. Shared backend failures stop the affected job rather than triggering a fresh answer draw. This contract makes the extension a test of selection among a given pool, with explicit failure behavior.

5 Experimental Setup

5.1 Seven-system baseline comparison

The formal v5 comparison uses M0 direct answering, M2 MedRGAG with the local all-Llama adaptation, M3 MA-RAG in its intrinsic configuration, M4 MedRAG–Llama, M5 MedRAG–Qwen, M6 MedGENIE version 2, and M7 TC-RAG with the selected typed-control version 2. M0, M2, M4, and M6 use Llama-3.1-8B-Instruct; M3, M5, and M7 use Qwen3-8B (Dubey et al., 2024; Yang et al., 2025). Thus, seven configurations span six baseline families and two underlying backbones. MA-RAG generates eight solver candidates per round for at most four rounds, with early stopping on agreement. MedRGAG, MedGENIE, and TC-RAG are the documented local adaptations, rather than claims of exact reproduction of every published model and resource setting.

Each selected configuration covers the same 13,905 unique inputs and 15,616 scoring mappings. Coverage combines new predictions with strictly matched historical reuse where available: M0 reuses 8,877 and generates 5,028 predictions; M2 reuses 6,687 and generates 7,218; M3 reuses 3,690 and generates 10,215. M4–M7 each have 13,905 new predictions in their selected complete run. Full coverage therefore means complete evaluated outputs, not seven simultaneous runs from scratch. Invalid and unaccepted terminal outputs remain in the common denominator.

The selected M4 baseline includes structured-output repair, whereas M5 retains its earlier unconstrained output protocol. Corpora, upstream procedures, and budgets also vary across families. Cross-system differences compare these concrete settings and cannot all be assigned to an isolated algorithm or backbone effect. M1, an earlier MedRGAG initial retrieval control, and the superseded i-MedRAG implementation are historical experiments, not additional completed v5 configurations (Appendix A).

5.2 Seven-system terminal-head evaluation

We evaluate the frozen terminal heads on every member of R1, R2, and R3 for all seven configurations. In R1, each configuration routes 1,780 diagnostic inputs to catalog re-decision and 72 prescribing inputs to rules, retaining the other 1,014 answers. R3 contains 680 catalog inputs, 10 rule inputs, and 43 retained single-choice answers. R2 covers 55 of 79 inputs with rules and retains the remaining 24. Uncovered inputs remain in their category denominators.

Diagnostic scores are generated with each configuration's actual original context and backbone; earlier completed scores and exactly identical requests are reused as recorded. The fact extractor sees the current question, options, and required fields, but no method-specific retrieved history or initial answer. Consequently, its two backbone-specific fact sets can be fixed and combined offline with seven original-answer sets. This is a valid comparison of seven terminal systems, while the rule branch is not seven newly executed upstream retrieval pipelines. No backbone is substituted to improve another system's head result.

We compare the diagnostic head with single-pass catalog selection and uncorrected voting. For R2, a constant "none" control intervenes on the same 55 inputs and preserves each original answer on the other 24. Remaining-support correctness is reported alongside its total. Additional same-facts execution, question-scope, and external-source evaluations are kept within their original experimental scopes. The original formal baseline scores and the development head scores remain separate tables.

5.3 Two-configuration CF-MoA composition and verification

The later six-arm experiment uses only the two MedRAG configurations M4 and M5, with Qwen thinking disabled. Within each configuration, all six arms share the same frozen full-v5 input scope and original contexts. It tests the composition of established heads with further operations, followed by additional candidate verification; it does not extend the six-arm experiment to all seven systems.

The ordinary F readout uses bfloat16 weights, temperature 0.7, top-p 1.0, a 2,048-token output budget, and a 131,072-token context limit under the saved vLLM 0.8.5 profiles. The F readout uses structured native output in both configurations. The existing catalog and fact-extraction components retain their own shorter-context, temperature-zero profiles. Exact per-component settings, prompts, code snapshots, and reuse bindings remain in the released experiment package. The experiments use inference and deterministic tools; no new model training is reported.

Table 1 summarizes the six evaluation arms.

Table 1: Six evaluation arms. C1 and B denote the same adopted operation core.

Arm	Decision rule	Role in the comparison
C0	Original MedRAG configuration	Original-system reference
C1 / B	Adopted operation core	Strong system baseline
C2	B plus candidate-only binary verification	Same-pool verification without saved rationales
C3	B plus rationale-conditioned binary verification	Main research extension
C4	B with old F votes plus three additional answers	Expanded-pool sampling control
C5	B plus joint selection over the same candidates and rationales	Alternative same-pool selection control

C4 uses the predefined supplementary seeds 42, 43, and 44 on eligible ordinary-F disagree-

ments. A failed required supplementary answer leaves the control unavailable, rather than silently converting it into a two-draw control. C5 scores a joint finite-code choice under two opposing code assignments. C2, C3, and C5 preserve the existing pool. Their comparisons share the original computation but have different additional request and prompt costs; none is described as equal-cost merely because a common budget ceiling exists.

Complete v5 is the scope of this later experiment. Earlier 121-input and 52-input development panels informed its design and are not independent confirmation sets. The older 52-input panel still lacks four original M5 C0 outputs; that historical gap does not affect the complete full-v5 C0 results reported here. No new model calls or native grading are performed for this manuscript; the figures and tables are rebuilt from frozen completed results.

5.4 Failure retention and computation accounting

All arms retain the complete input and native-unit denominators. An unavailable answer counts against performance under the frozen contract and is also reported explicitly. We separate a valid but incorrect answer, an unavailable answer, and a verifier failure with a retained F answer. Pool-trigger coverage includes the full scope and is not filtered by subsequent success or gold correctness.

For the later composition experiment, per-method computation assigns the shared B computation once to each independently considered B-based method, then adds that method's own incremental requests. This estimates the attributable computation of a method assembled from the recorded components. It is distinct from the physical ledger for the multi-arm experiment, where shared calls were actually executed once. Exact cache reuse is not treated as free historical computation. The upstream C0 and retrieval costs, and some historical event times, remain unknown. Reported event seconds are not exclusive GPU hours, end-to-end latency, or energy measurements.

6 Results

6.1 Seven systems reveal different support-update weaknesses

Table 2 reports the complete selected baseline comparison. ALL means range from 47.35% to 51.61%, while R1, R2, and R3 expose substantially lower and more variable scores. MA-RAG has the highest ALL point estimate (51.61%) and R5 point estimate (73.37%), yet the lowest R3 score (8.59%). MedRAG–Qwen leads R2 at 41.77% but scores 25.24% on R3; MedRAG–Llama reaches 30.42% on R3. These observations motivate task-specific analysis rather than treating the overall ranking as a ranking of all reasoning capabilities. We do not claim statistical superiority from these point estimates alone.

The source composition helps explain what needs investigation. R3 contains 680 MedEinst diagnostic units, 10 prescribing units, and 43 single-choice units. R2 contains 75 MedPIC units, of which 66 have a "none" answer and nine still require a drug option, plus four single-choice units. R1 includes 1,780 diagnostic, 788 evidence-relation, 96 prescribing, and 202 single-choice units. Low scores across these strata therefore do not identify one common failure mechanism. In particular, many diagnostic failures are unrecognized by the frozen label mapping, while R2 improvements must distinguish withdrawal from retaining another valid support path.

6.2 Terminal heads improve all seven configurations

Table 3 and Figure 1 give the complete original-to-head comparison. All 21 configuration-by-stratum comparisons have positive net gains. R1 increases by 14.83–19.12 percentage points, R2 by 16.46–29.11 points, and R3 by 18.69–35.20 points. For example, MedRAG–Llama improves

Table 2: Formal v5 results for seven system configurations spanning six method families. Values are percentages of native unit scores. ALL uses 6,104 unique units; category sets overlap. Each configuration covers 13,905 inputs. L: Llama-3.1-8B-Instruct; Q: Qwen3-8B. M4 uses the repaired structured output; M5 retains its original output configuration. Corpora, retrieval procedures, and inference budgets also differ, so these are descriptive system comparisons.

System	LM	ALL (6,104)	R1 (2,866)	R2 (79)	R3 (733)	R4 (13)	R5 (3,600)
M0 Direct	L	48.10	30.04	8.86	17.87	38.46	65.70
M2 MedRGAG	L	48.81	31.33	7.59	24.69	53.85	65.88
M3 MA-RAG	Q	51.61	26.73	34.18	8.59	38.46	73.37
M4 MedRAG–L	L	48.20	27.84	29.11	30.42	53.85	65.28
M5 MedRAG–Q	Q	47.35	28.58	41.77	25.24	53.85	63.42
M6 MedGENIE	L	48.02	29.94	18.99	21.42	61.54	65.15
M7 TC-RAG	Q	49.43	31.44	29.11	28.92	30.77	66.17

Table 3: Archived original → head correct counts on R1–R3 for all seven configurations. The head combines candidate re-decision with rule support and preserves original answers outside coverage. These are analyses of exposed development data, using saved method-context decisions and shared backbone-level rule facts; they are not seven new upstream pipeline reruns. R3 is contained in R1; the three sets contain 2,941 distinct units.

System	R1 ($n = 2{,}866$)	R2 ($n = 79$)	R3 ($n = 733$)
M0 Direct	$861 \to 1,329$	$7 \to 29$	$131 \to 338$
M2 MedRGAG	$898 \to 1,323$	$6 \to 23$	$181 \to 340$
M3 MA-RAG	$766 \to 1,260$	$27 \to 43$	$63 \to 238$
M4 MedRAG–L	$798 \to 1,225$	$23 \to 39$	$223 \to 386$
M5 MedRAG–Q	$819 \to 1,367$	$33 \to 46$	$185 \to 443$
M6 MedGENIE	$858 \to 1,317$	$15 \to 38$	$157 \to 305$
M7 TC-RAG	$901 \to 1,331$	$23 \to 37$	$212 \to 349$

from 798 to 1,225 correct R1 units, 23 to 39 R2 units, and 223 to 386 R3 units; MedRAG–Qwen improves from 819 to 1,367, 33 to 46, and 185 to 443, respectively. The same frozen decision policy is used across all configurations. These are established development gains from changing the terminal decision procedure, including in systems whose upstream procedures differ substantially.

The R1 extension adds evidence beyond simply recounting the earlier R3 result. Among the 2,133 R1 units outside R3, the seven configurations gain 261–319 correct answers. Its active new scope includes 1,100 diagnostic inputs and 62 additional rule-covered inputs. Conversely, the 1,014 uncovered R1 inputs retain their original answers, including the 450 selected R5 units that also belong to R1. This establishes gains over the complete declared R1 scope without implying that every R1 task was modified.

Gains coexist with regressions. Across the complete R1 stratum, M4 repairs 481 errors and harms 54 previously correct answers; M5 repairs 584 and harms 36. For R3 the corresponding counts are 201/38 and 278/20. R2 M7 improves overall while its remaining-support subset declines from 3/9 to 2/9. The system-level result is therefore consistent positive net improvement across seven configurations, rather than universal per-case or per-subgroup improvement.

The saved baseline and head records also agree across shared categories: the R1–R3 intersection reproduces all 5,131 configuration-by-input outcomes. This consistency supports the reuse accounting; it does not turn overlapping observations into additional independent evidence.

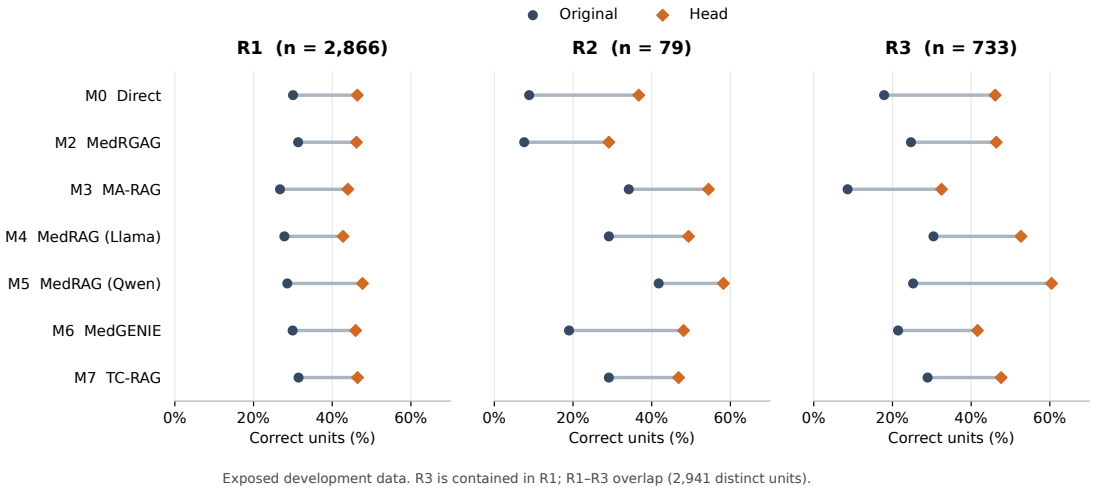


Figure 1: Original and terminal-head accuracy for all seven configurations in the complete R1, R2, and R3 development strata. Every uncovered input remains in its stratum's denominator. R3 is a subset of R1, and the panels share cases and backbones; they do not represent three independent test sets or seven complete CF-MoA evaluations.

6.3 Controls locate the gains and their limits

Table 4 separates the full-head effect from its components. On the 680 R3 diagnostic inputs, mean accuracy across configurations rises from 20.97% for original answers to 45.44% for single-pass catalog selection, 46.47% for uncorrected six-vote selection, and 46.13% for corrected six-vote selection. Thus, catalog re-decision already obtains most of the observed gain. Correction changes correct counts relative to uncorrected voting by 0, −10, +4, 0, +8, −12, and −6 in M0/M2/M3/M4/M5/M6/M7 order. The full component is retained as one frozen policy; we do not choose each system's best aggregation after observing these results.

Of 1,272 R3 diagnostic repairs across systems, 1,186 start from an unmapped diagnosis, 23 from another native-invalid answer, and 63 from a valid wrong answer. Unmapped outputs include names, granularity, lists, and incorrect diseases, so the first two groups' 95.05% share is not a clinical adjudication of "format-only" repairs. It does show why a large gain under the frozen mapping cannot all be attributed to better medical inference. Earlier matched 40-input controls provide a narrower distinction: M4 and M5 respectively obtain 12 and 9 correct answers out of 40 originally, 15 and 11 after name normalization alone, and 18 each after reading the patient and choosing within the same catalog. Re-decision has value beyond that presentation control on this development subset.

For R2, all 130 repairs start from native-valid original answers and use exact-set scoring. A policy that selects "none" on the same 55 covered inputs and preserves the other 24 original answers nevertheless scores 48, 47, 58, 58, 56, 52, and 54 out of 79, above every default head total. It gets 0/9 remaining-support cases correct, whereas the head gets 2–6/9. Both findings matter: the head changes support behavior, but its aggregate score has not surpassed this strong answer-distribution control. With final facts fixed, switching from all relevant rules to question-scoped rules improves both backbones from 24/48 to 48/48 on controlled scope probes. Those probes test a specific execution distinction, not 48 independent clinical cases.

Table 4: Controls expose limits of attributing the head gains. (a) R3 diagnosis subset ($n = 680$): original, one catalog order (order 0), majority across catalog orders without calibration, and calibrated majority. Calibration does not improve every system. (b) R2 ($n = 79$): the same-coverage None control predicts None on 55 covered inputs and preserves 24 original answers. Its higher aggregate score reflects 46 covered None targets; it misses all nine covered cases with remaining support. The last two columns isolate these nine cases.

(a) R3: correct out of 680

System	Original	Order 0	Majority	Calibrated
M0 Direct	116	299	314	314
M2 MedRGAG	163	320	325	315
M3 MA-RAG	33	215	200	204
M4 MedRAG–L	203	339	359	359
M5 MedRAG–Q	160	382	402	410
M6 MedGENIE	137	296	288	276
M7 TC-RAG	186	312	324	318

(b) R2: aggregate and remaining-support subset

System	Head /79	None /79	Head /9	None /9
M0 Direct	29	48	6	0
M2 MedRGAG	23	47	4	0
M3 MA-RAG	43	58	3	0
M4 MedRAG–L	39	58	4	0
M5 MedRAG–Q	46	56	6	0
M6 MedGENIE	38	52	5	0
M7 TC-RAG	37	54	2	0

6.4 External component checks bound the transfer claim

Frozen component checks were completed in addition to the exposed v5 development experiments. On 47 qualifying ASCENT source records,[1] comprising 42 source-training and five source-test records, the catalog head obtains 38 and 39 correct answers for Llama and Qwen under the frozen canonical-name metric, exactly the totals of same-prompt single-pass catalog selection. These records were subsequently reused for targeted development; later variants cannot treat them as unseen confirmation data. On three successive CUPCase batches totaling 18 source records,[2] corrected voting obtains 15 and 16 correct answers out of 18 for Llama and Qwen, uncorrected voting obtains 16 and 17, and single-pass native four-candidate selection obtains 15 and 18. These are finite-source adaptations using two backbones, not seven upstream retrieval systems or an independent counterfactual paired test. They support the operability of the candidate interface while leaving stable additional calibration benefits unestablished.

In a separate frozen rule evaluation, 11 constructed cases each supply a base, a support-added, and an irrelevant-information endpoint. The program obtains 21/33 and 25/33 correct, compared with 6/33 and 17/33 for a model executing the same facts, rules, scope, and assembly contract. This is evidence for finite rule execution under the tested resources. It leaves errors in fact extraction and UNKNOWN inheritance, and it does not supply a new independent clinical prescribing cohort. A 201-question relation control, not included in the seven-system

[1] Source implementation and data: https://github.com/naver-ai/ascent/tree/89091716698d6a2fa78b81d8380e7fec57e73fa5.

[2] Author dataset: https://huggingface.co/datasets/ofir408/CupCase/tree/157a85b1a0fa6d0e25e5209a1cdaf41cf1ff8753.

head, improves from ordinary redo to target-scoped redo by $102 \rightarrow 118$ for M4 and $120 \rightarrow 122$ for M5. The respective review-cluster intervals are $[+1.03, +14.93]$ and $[-5.19, +7.37]$ percentage points; only the first confirms an increment within that exposed scope. These controls preserve successful component results while identifying which stronger claims remain unsupported.

6.5 The later CF-MoA core improves the full native-unit mean

The remaining results concern the separate M4/M5 six-arm composition experiment. Table 5 reports every arm with both native-unit and input-level outcomes. B increases M4's ALL mean from 48.1990% to 58.2924%, and M5's from 47.3487% to 59.3616%. The paired family intervals for these gains are $[+8.9580, +11.2340]$ and $[+10.9177, +13.1510]$ percentage points, respectively. These are overall effects of the adopted components and their additional resources and computation. They do not isolate the contribution of the later verification stage.

Figure 2 displays the native-unit point estimates.

Table 5: Full-v5 results. Each row contains 13,905 inputs, 15,616 scoring mappings, and 6,104 non-reference native units. ALL is the mean of native-unit scores. Unavailable inputs remain in the denominators.

Config.	Arm	ALL (%)	Correct inputs	Unavailable inputs
M4 Llama	C0	48.1990	7,441	0
M4 Llama	B	58.2924	8,924	51
M4 Llama	C2	55.9939	8,554	51
M4 Llama	C3	56.4815	8,649	51
M4 Llama	C4	58.3219	8,951	157
M4 Llama	C5	56.7295	8,660	51
M5 Qwen	C0	47.3487	6,913	0
M5 Qwen	B	59.3616	8,971	22
M5 Qwen	C2	59.3562	8,843	22
M5 Qwen	C3	60.2889	9,060	22
M5 Qwen	C4	59.6723	8,878	86
M5 Qwen	C5	58.9690	8,739	22

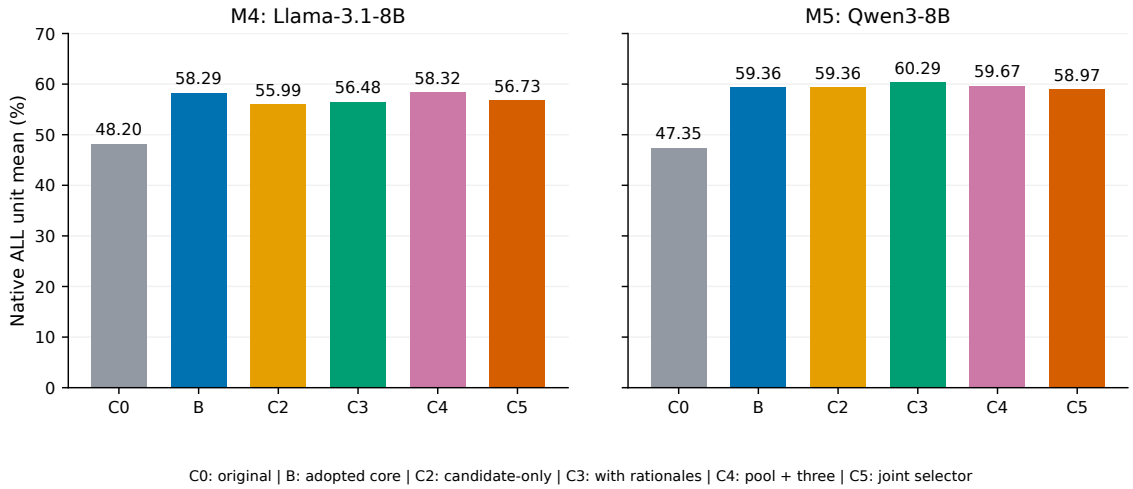


Figure 2: Full-v5 native-unit scores for the six arms. All bars use the same 6,104-unit denominator; the chart shows point estimates, while paired contrast uncertainty is reported separately.

Input correctness and native-unit means need not rank arms identically. For example, M5 C2 has fewer correct inputs than B, while their ALL means are nearly equal. Different native-unit structures and shared mappings explain why an input-level count cannot substitute for the declared main metric. We retain both views rather than selecting the more favorable one.

6.6 Additional verification has configuration-dependent effects

Table 6 reports the paired contrasts, and Figure 3 displays their bootstrap intervals.

Table 6: Paired contrasts in the ALL mean. Effects and 95% family bootstrap intervals are in percentage points. These are descriptive intervals on exposed development data.

Config.	Contrast	Effect	95% interval
M4	B − C0	+10.0934	[+8.9580, +11.2340]
M4	C3 − B	−1.8108	[−2.5287, −1.0819]
M4	C3 − C2	+0.4877	[−0.1275, +1.0907]
M5	B − C0	+12.0129	[+10.9177, +13.1510]
M5	C3 − B	+0.9273	[+0.3039, +1.5562]
M5	C3 − C2	+0.9327	[+0.4641, +1.4010]

C3's effect relative to B is negative for M4 and positive for M5. At input level, M4 C3 repairs 436 B errors but damages 711 B-correct inputs, for a net loss of 275 correct inputs. M5 repairs 452 and damages 363, for a net gain of 89. Both configurations therefore contain repairs and regressions. The aggregate difference is not explained by counting successful repairs alone.

Including rationales improves the point estimate relative to C2 in both configurations. The M4 interval crosses zero, whereas the M5 interval is positive. This comparison isolates the saved-rationale condition within the verifier's fixed candidate-pool design; it does not establish that the rationales are factually reliable. For M4, improving on C2 still leaves C3 below B.

C4 has positive ALL point differences from B in both configurations, but the broader results do not establish a consistent advantage. M4 C4 is only +0.0295 points above B, with an interval of [−0.5445, +0.5895], and has 106 additional unavailable inputs. M5 C4's ALL mean is above B, but its input-level correct count is lower and unavailable inputs increase by 64. C5's ALL point estimate is below B in both configurations. We preserve C3 as the declared research extension and B as the adopted default, without selecting a different winning arm for each configuration.

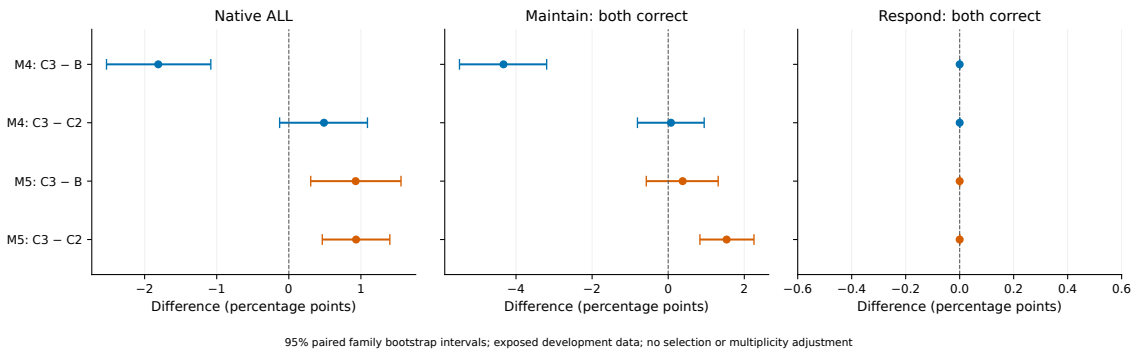


Figure 3: Paired family bootstrap contrasts. Zero marks no change; intervals quantify exposed-benchmark variation and do not provide selection-adjusted or independent-confirmation evidence.

6.7 Source-level gains and regressions coexist

The aggregate C3 effect contains substantial source variation. On MedCounterFact (M20), C3 improves over B by +2.4112 points for M4 and +9.6447 for M5. On the AMQA-derived source (M12), the corresponding differences are −5.8219 and −2.0548. Cultural-cue questions (M11) also decrease in both configurations, by −1.5474 and −3.8685 points. These are source-specific point estimates; we do not interpret a ranking of small source effects as a corrected significance test.

The 13-unit R4 clinical source illustrates the danger of suppressing unfavorable strata. B's M5 score is already below C0, and C3 recovers only one unit relative to B. M4 C3 instead loses three units relative to B. The source remains in the full score. The system's separate explicit-model component has no clinical-v5 coverage, so these outcomes cannot be presented as evidence that model-based consequence execution succeeded or failed on these inputs.

M01 diagnosis scores are identical across B and C2–C5 because the extension leaves the catalog route unchanged. Other unchanged source scores should not automatically be interpreted as evidence of a particular routing mechanism: multiple routes or changed answers can produce the same source aggregate. The released format and route tables provide the needed separation. Figure 4 and Appendix B retain every source rather than reporting a selected improvement subset.

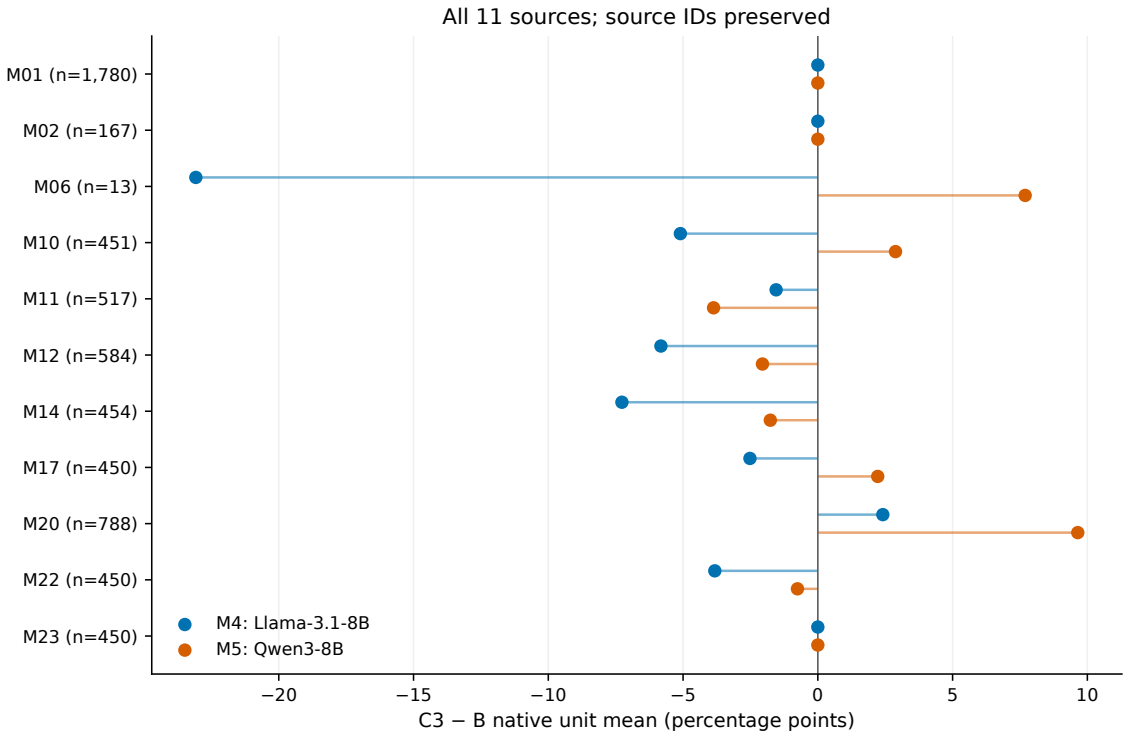


Figure 4: C3 minus B for every source and configuration. Sources are retained in identifier order, including the small R4 clinical source; no source weighting replaces the primary ALL score.

6.8 Paired outcomes distinguish maintenance from response

Table 7 reports the both-endpoints-correct counts.

Relative to B, C3 reduces M4 maintain correctness by 296 pairs and increases M5 by 26. The M4 change is −4.3288 percentage points, with a family interval of [−5.4864, −3.1947]. All

Table 7: Both-endpoints-correct counts. Maintain uses 6,838 pairs; respond uses 1,780 pairs. Equal respond counts are an observed result of the present outputs, not evidence that the verifier solves arbitrary changes.

Arm	M4 maintain	M5 maintain	M4 respond	M5 respond
C0	4,005	3,528	13	27
B	4,440	4,460	347	360
C2	4,139	4,381	347	360
C3	4,144	4,486	347	360
C4	4,509	4,412	347	360
C5	4,150	4,246	347	360

B-based arms have the same respond count in each configuration. The core's improvement over C0 on respond pairs is therefore retained, but additional verification has no observed aggregate increment on this fixed subset. Pair correctness measures answer adequacy at both endpoints and should not be reduced to a preference for either changing or keeping an answer.

6.9 Both support-update operations contribute within the rule subset

The operation ablation reuses the same extracted facts, support states, and native assembly policy for all 123 rule-routed inputs. Disabling an operation changes assembly alone; it does not rerun extraction or select a favorable subset. Table 8 therefore tests the role of the two update directions within the adopted shared rule core.

Figure 5 displays the corresponding ablation counts.

Table 8: Existing support-operation ablation. All cells use 123 inputs and 123 native ALL units. Unavailable counts are one for M4 and 14 for M5 in every arm.

Config.	Full B correct	Disable addition	Disable removal
M4	69	43	53
M5	71	48	60

Removing addition lowers correct counts by 26 and 23; removing withdrawal lowers them by 16 and 11. Each ablation can also repair some individual errors, so these net changes do not imply that either operation is always correct. The results support retaining both directions in this shared execution policy. They do not measure an interaction between two independently trained agents or an ablation of all 13,905 inputs.

6.10 Additional selection incurs measurable computation

Ordinary F covers 9,293 inputs per configuration. At least two observed legal keys occur on 2,964 M4 inputs and 2,295 M5 inputs, corresponding to 21.3161% and 16.5049% of the complete 13,905-input scope. These coverage counts include unavailable cases and invalid old votes in their accounting; they are not counts of successful verification. Candidate pools depend on the model configuration, so the triggered subsets are not identical controlled populations across configurations.

Table 9 reports the attributed B and C3 request and token counts.

C3 adds approximately 22.25% to B's attributed request count for M4 and 16.84% for M5. Additional computation accompanies a negative quality change in one configuration and a

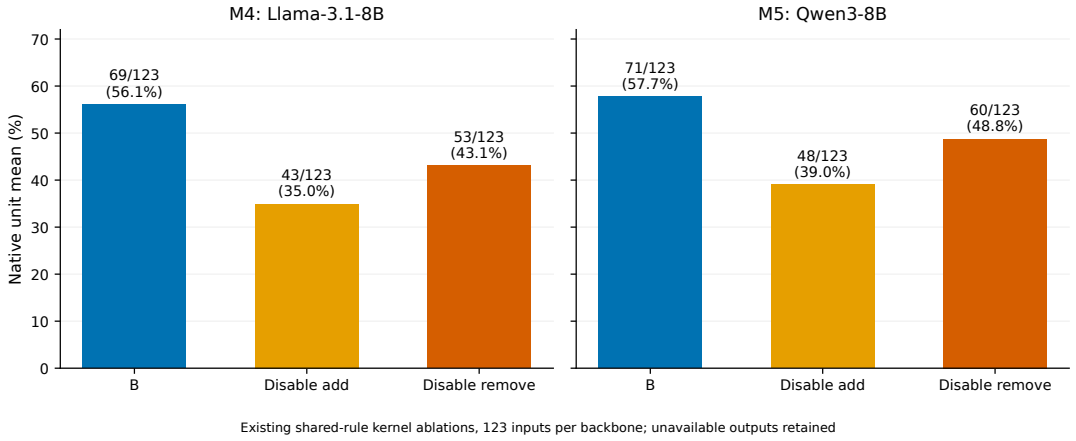


Figure 5: Correctness of the shared rule component with addition or removal disabled. The denominator is the complete 123-input rule subset; the same facts and support states are reused.

Table 9: Attributed request costs for B and C3. The table includes the shared core once per method and excludes unknown common C0/retrieval costs. Tokens are the total attributed C3 tokens, not only verifier tokens.

Config.	B requests	C3 extra requests	C3 total requests	C3 input tokens	C3 output tokens
M4	59,800	13,308	73,108	358,643,985	9,699,532
M5	58,301	9,820	68,121	294,927,306	7,655,025

positive change in the other. Figure 6 displays the recorded B-based arms against their method-attributed requests. C0 is not assigned zero cost to place it on this frontier because its common upstream cost is unknown.

The physical ledger for this completed multi-arm full-v5 run records 162,979 requests, 957,810,324 input tokens, and 18,291,906 output tokens. These totals must not be divided by six to estimate a method cost, and method-attributed totals must not be summed to recover the physical ledger because they share B. They also do not represent the total cost of the project's earlier seven-system, head-development, or external-validation experiments. For each configuration and B-based method, 2,503 inputs have historical timing gaps. The analysis therefore reports request and token attribution with missing-time markers, not a fully measured end-to-end efficiency claim.

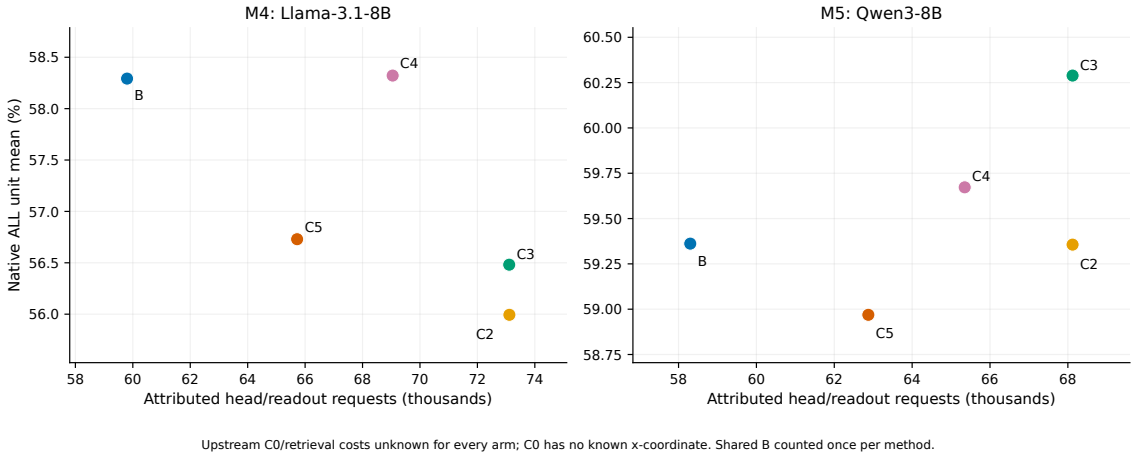


Figure 6: Native-unit quality versus attributed physical requests for the B-based methods. Shared B computation is assigned once to each method; unknown common original-system and retrieval costs are excluded. Lines or proximity do not imply equal prompt length or equal latency.

7 Discussion

The seven-system results establish that terminal decision procedures can improve a range of medical question-answering systems under a common native evaluation contract. The benefit is not confined to the two MedRAG configurations: direct answering, MedRGAG, MA-RAG, MedGENIE, and TC-RAG also improve throughout the three declared development strata. Preserving these results alongside their controls is necessary to understand the research outcome. Mixed contributions from calibration, or incomplete external validation, do not erase the measured gain of the complete terminal head.

What changes depends on the task. The diagnostic component provides an explicit answer space and a new original-model decision; the rule component executes support conditions and combines them with the original answer. These are distinct interventions, and R1's reuse of R2/R3 components demonstrates useful transfer across support requirements without constituting three independent algorithms. A common component can also preserve method differences through the original context, UNKNOWN inheritance, and fallback. The evidence supports system improvement with explicit attribution limits, rather than a claim that all failures share one cause.

The later CF-MoA core combines the catalog, rules, and repeated readout in two configurations. Its improvement concerns that broader composition and its additional computation. It neither replaces the seven-system findings nor extends the full CF-MoA result to five untested configurations. The composition's gain also cannot be reassigned to candidate verification introduced afterward.

Additional candidate verification poses a distinct selection problem. By design, C3 cannot recover an answer key absent from its pool, and it can replace a correct selected answer with an incorrect existing candidate. The repair and damage counts establish that successful and harmful switches both occur; they do not measure the frequency of missing correct candidates or diagnose the medical cause of each mistake. Pool quality and selector quality should therefore be measured separately. Enlarging the pool is also not automatically beneficial: the C4 control changes both candidate opportunity and failure exposure, with its unavailable outputs retained in the result.

Rationales provide extra information to the verifier, but they can contain unsupported statements or persuasive explanations for a wrong key. Their positive incremental point estimates

over C2 justify further study of how they affect selection. They do not certify rationale truthfulness, establish calibrated confidence, or guarantee improvement over majority-based readout. A direct evaluation of rationale validity and an independent replication of pool generation would be needed to test those stronger interpretations.

The source and pair analyses constrain the use of an aggregate gain. A positive overall score can coexist with harm on a particular source, and a changed overall mean can coexist with unchanged respond-pair performance. No arm was selected separately for each source or backbone after inspecting these outcomes. Keeping the adopted core as the common default preserves the distinction between an observed local gain and a deployment or generalization decision.

Finally, the five-role description is useful only when tied to actual computation. Shared rule addition and removal are two functions of one execution. A model-based executor with zero coverage cannot explain a clinical benchmark gain. Recording these facts makes the architecture interpretable without implying that every question benefits from five-agent collaboration.

8 Limitations

Development exposure and confirmation. The full v5 benchmark was already exposed and contributed to baseline repair, terminal-head development, and later composition selection. Current bootstrap intervals do not remove that exposure. Frozen component evaluations on ASCENT, CUPCase, constructed rule cases, and new physiological profiles were completed and retain their stated narrow scope. ASCENT's 47-record check combines source-training and source-test records and was subsequently exposed to further development; only its initial frozen comparison retains that evaluation status. A planned new-family evaluation of the final CF-MoA composition and a second independently generated F pool were deferred and have not been executed. The final composition therefore lacks independent confirmation and across-pool replication; this does not reclassify the completed component checks as unexecuted.

Scope and scoring. The sources differ in task semantics, patient realism, perturbation construction, and reference quality. R labels overlap, some groups are small, and the 13-unit R4 clinical subset has especially limited coverage. Classification follows the project's documented target-level review rules; no clinician inter-annotator agreement estimate is reported. Canonical diagnostic matching can penalize clinically related wording, while source expected answers do not provide a new clinical adjudication. An empty list of newly established data defects in the final closeout is not evidence that all items and labels have undergone a complete expert medical review.

Attribution and model comparison. Seven configurations share two backbones and six method families, and their upstream resources and budgets differ. Rule-head results reuse backbone-specific facts with seven initial-answer sets, while diagnostic re-decision uses each configuration's actual context. Neither procedure establishes seven independent data replications or seven new complete upstream runs. B combines already developed components with additional resources and inference. Specified local comparisons, such as fixed-fact rule ablations and C3 versus C2, isolate narrower design changes; single-pass and repeated catalog decisions have different call budgets. M4 and M5 differ in output protocol as well as model. Their disagreement in C3's effect is configuration dependence; a pure backbone explanation requires matched-protocol experiments. Candidate-trigger membership also differs because each model generates its own pool.

Component coverage. The rule head covers a limited public rule bank and fixed question-scope patterns, and can inherit errors through UNKNOWN states. R2's answer imbalance favors the constant "none" control. The diagnostic catalog is finite, and the small external panels do not establish stable calibration gains. R4's executor requires a complete supplied model; the

later R5 learning candidates were not adopted. The seven-system R1–R3 head table does not establish seven-system R4/R5 optimization.

Clinical applicability. This is an offline benchmark study, not a prospective patient study, treatment trial, or deployment evaluation. Rule coverage is limited, and the A4 physiological results concern a supplied mathematical model. Neither exact-match improvements nor successful model execution establish clinical safety or correctness outside those contracts.

Computation and reproducibility. Some original-system costs and historical timings are unknown. Shared-call attribution describes the measured components rather than complete end-to-end resource use. The figure package reproduces reported summaries and can check them against the released compressed result records; it does not rerun the full language-model experiment. Original input data, licenses, model access, retrieval indices, and environment-specific paths remain separate prerequisites for fresh inference.

9 Conclusion

We evaluate seven medical question-answering configurations on a common full-v5 benchmark and improve every configuration on the complete R1, R2, and R3 development strata through question-scoped support recomputation and original-model catalog re-decision. Component controls distinguish these system gains from the variable effects of calibration, repeated decisions, and answer-distribution policies. The subsequent two-MedRAG CF-MoA experiment evaluates a broader composition and finds strong core gains, followed by mixed incremental effects from rationale-conditioned verification. The seven-system comparison, component studies, and six-arm follow-up answer complementary questions; each retains its actual scope and computation. The results support reusable terminal decision components with measured development benefits, while independent confirmation of the final composition and candidate-pool replication remain future work.

Declarations

Data and code availability. The paper delivery contains native LaTeX/BibTeX, frozen summaries, source hashes, plotting/build scripts, and checks distinguishing original generation from reuse. Composition code, configurations, and results are in the full-v5 release and the transparent closeout release (commits `d857934` and `2d56c71`). The repository is private and requires authorized access. Source datasets retain their original access and licensing conditions.

Ethics declaration. This study reuses benchmark material and historical outputs, with no new patient recruitment or clinical intervention. Institutional exemption or approval is not established here; authors must verify applicable requirements, source-use permissions, and approval identifiers before submission. Full patient-like prompts and private execution traces are not included in the figure package.

Author contributions. Author names, affiliations, and contributor-specific Contributor Roles Taxonomy (CRediT) roles have not been supplied for this draft and must be completed by the authors. No authorship is assigned to the writing tools.

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AI assistance. Codex, with the academic-paper workflow, assisted with evidence organization, reference checks, English drafting, and rendering code. No new model-evaluation results were generated for this manuscript. Authors remain responsible for review and any venue-specific disclosure.

A Experimental Development and Scope Boundaries

The reported system was developed through several experiments with different input contracts. The following account identifies the decisions that connect the main results; the released records retain the detailed configurations, failures, and intermediate outputs. Historical scores are not merged into the current v5 leaderboard.

Early replication and paired-input prototypes. Early work included direct answering, MedRGAG reproduction and local adaptations, a world-model proxy, and Delta/Residual Adapter variants. The apparent advantage of world-model sorting did not survive a matched prompt-and-input comparison: both it and blind direct sorting scored 64/85. A subsequent counterfactual experiment identified a label-oracle condition; that condition was rejected as valid method evidence, and the corrected label-free version did not establish the proposed mechanism. A later Residual Adapter achieved 80.0% with MA-RAG on a 500-pair four-choice task, while a separately trained without-MA configuration reached 81.8%. Those experiments provided useful controls, but their paired-input and constructed-choice permissions differ from the current independent-endpoint native protocol. They neither establish synergy nor supply extra v5 baseline scores.

Baseline lineage. The evaluation progressed from earlier source-level and v2/v3 comparisons to the frozen v5 selection. M1 denotes MedRGAG's initial retrieval control, not a separate MedRAG family; its earlier completed evaluations do not provide a full v5 result. Historical M6 denoted i-MedRAG. Its original v5 queue paused after 52 of 13,905 inputs; a later 128-input performance check was not a completed v5 evaluation. The selected current M6 is MedGENIE and preserves a different method identity. M4's structured-output repair and M7's typed-control revision are included in the selected baseline versions, with earlier failures retained in their own records.

From errors to terminal heads. R2 development first tested ordinary revision, free-form support audits, local withdrawal, atomic decisions, additional guideline context, and longer reasoning. Their failures led to a narrower division of labor: the model extracts current facts, and a fixed rule executor handles conditions, exceptions, question scope, and independent support paths. An answer-assembly type error was repaired using saved outputs, and its earlier negative conclusion was withdrawn rather than retained as a method failure. The final default preserves UNKNOWN initial choices and records its remaining-support regressions.

R3 first tested evidence audits, query changes, and training-case memory. A high-scoring nearest-neighbor head was withdrawn from the core method because it returned retrieved training labels instead of making the intended original-model decision. The subsequent public-catalog head is a distinct implementation. Patient presentation, catalog selection, permutations, and prior correction were evaluated separately. More elaborate pairwise comparisons, observation selection, knowledge allocation, relation counting, and NLI candidates did not replace the retained head. R1 then extended the frozen diagnostic and rule components beyond R3, with complete seven-system accounting. Single-choice extension and revised fact extraction had negative results; a relation candidate remained a separate two-configuration experiment.

R4: conditional consequence execution. The original 13-unit HIV stratum has seven-system baseline results, but subsequent R4 head development used two backbones. Clinical prompt and evidence-binding variants did not establish a retained clinical head. Formal and biological tasks informed the later implementation, whose adopted scope is explicitly supplied physiological models and states. Frozen validation on 24 new virtual profiles, each with 12 questions, is reported in Appendix D. It is completed conditional model validation, not seven-system optimization of the 13 HIV units or evidence of unrestricted clinical causal reasoning.

R5: retained readout and unadopted learning candidates. The historical F component combines two NLI paths and two label mappings, or up to five native-answer draws for its other supported tasks. Its 208-unit development/validation panel contains 64 initial devel-

opment units, 128 extension units, and 16 new groups evaluated after freezing. Across the 208 units, weighted paired correctness changes from 60.19% to 65.90% for Llama and from 59.17% to 66.47% for Qwen. Within the 16 frozen-after groups, both-correct pairs improve from 11 to 12 for Llama and from 12 to 13 for Qwen relative to matched single answers. These limited results support retaining F as a component. Later support audits, learned representations, and fallback gates failed their controls and were not adopted; a further five-head training proposal remained unexecuted. They are not extra successful R5 methods hidden within the final composition.

Composition and verification follow-up. CF-MoA subsequently tested global aggregation and additional support, comparison, and maintenance variants. These were executed experiments with negative or mixed results, not merely unimplemented plans. A minimal wrapper then preserved the established B operations, while ordinary fresh answer generation and finally candidate verification were tested as additional interventions. Earlier 121-input and 52-input panels informed that process. The seven-system component migration replay demonstrated equivalence of saved results; its cached callbacks were not new quality measurements. The final six-arm, full-v5 experiment applies only to M4 and M5. Its 162,979-request physical ledger covers that run, not the total cost of the development history summarized here.

B Source Composition and Complete Reporting

Table 10 identifies the frozen source strata and their native ALL-unit counts. The cited artifacts identify upstream resources; the local inclusion and qualification rules remain those of the frozen v5 construction. In particular, dataset-card counts and original-paper sample counts must not be substituted for this table's evaluation denominators.

Table 10: Frozen source composition. Counts are non-reference native ALL units, totaling 6,104.

ID	Source artifact	ALL units
M01	MedEinst (Chen et al., 2026)	1,780
M02	MedPIC-Bench (MedPIC-Bench)	167
M06	HIV-treatment causal scenarios (Prosperi et al., 2026)	13
M10	CPV-MedQA (Benkirane et al., 2025)	451
M11	Counterfactual Cultural Cues (Rezaei and Shakeri, 2026)	517
M12	FairMedQA / AMQA (Xiao et al., 2025)	584
M14	DiversityMedQA (Rawat et al., 2024)	454
M17	MedPerturb (Gourabathina et al., 2025)	450
M20	MedCounterFact (Mo et al., 2026)	788
M22	MedRGB (Ngo et al., 2024)	450
M23	BioRAB (Li et al., 2025)	450
Total	Non-reference native units	6,104

The CPV citation describes the perturbation method; M10 uses the authors' MedQA extension, rather than substituting the paper's primary question cohort. MedPIC-Bench is cited through its available dataset card, with no inferred publication date. Version-specific reference metadata and source-to-stratum mappings are recorded in the companion bibliography audit.

MedEinst derives its diagnostic material from DDXPlus (Fansi Tchango et al., 2022); that upstream provenance is retained alongside the MedEinst citation.

Complete machine-readable tables accompany the draft: seven selected baseline configurations across ALL and R1–R5; all 21 original-to-terminal-head comparisons; the catalog and constant-answer controls; and, for the later CF-MoA study, all six arms by configuration; all 11 sources; overlapping R1–R5 categories; all five input formats; all maintain/respond pair scores;

every recorded family interval; and attributed costs and coverage. The source-level plot is a view of those complete tables. It does not define a new filtered benchmark. Input-format scores use input denominators, whereas source and R-category scores use native-unit denominators.

C Selection Procedure and Reproduction

The later CF-MoA core and verifier can be summarized as the following procedure. The source implementation, including native-interface special cases and error handling, remains authoritative.

```
Input: legal current input x; saved original context and answer
route = fixed_visible_capability_predicate(x)
B = execute_or_reuse_adopted_component(route, x)
if route is not ordinary F, or B is unavailable:
    return B
C = distinct legal keys actually observed in B's saved F pool
if |C| < 2:
    return B
for each candidate k in native order:
    choose earliest complete saved representative for k
    score correctness under both opposite A/B code mappings
    normalize each two-code distribution; average correctness scores
if verification has a technical failure:
    record failure and return saved F output
select maximum; exact ties prefer B, then native order
return the selected complete saved representative
```

The C2 procedure omits saved rationales from each verification request. C3 includes them as untrusted candidate reasoning. C5 replaces per-candidate binary scoring with joint finite-code selection. C4 performs three predefined additional native-answer draws and votes over old and additional answers; its failure policy is separate from verifier fallback.

To rebuild the experimental figures and tables, the companion package provides `scripts/build_figures_tables.py`. It reads frozen summary inputs and writes CSV, LaTeX tables, and PDF/PNG figures. Its optional raw-record check independently reaggregates the original compressed result records without invoking a model or grader. For the seven-system additions, `scripts/build_scope_figures_tables.py` reads the separate frozen aggregate inputs and checks 42 baseline values and 21 head comparisons before rebuilding three tables and the comparison figure. The companion audit receipts document the preceding full-record reconciliation of saved R1/R2/R3 head outcomes with the selected original scores, including source hashes. The shipped plotting script does not repeat that raw-record audit. Both operations aggregate existing outcomes; neither generates answers, runs a semantic grader, or supplies independent validation. The manuscript is maintained directly in `latex/main.tex` and its included section files, with citations resolved from `references.bib` by BibTeX. The accompanying `latex/build.sh` compiles this source with a TeX engine; the normal document build does not require Markdown, Pandoc, or Typst. Source and output hashes and execution receipts identify the exact inputs used. A rebuild of these artifacts establishes agreement with the saved results, not independent experimental replication.

D Historical Explicit-Model Validation

The adopted A4 component was previously evaluated on 24 virtual parameter/initial-state profiles, with 12 questions per profile and model configuration. The 288 questions per configuration use one fixed simglucose model, four fixed actions, and three question types. The software snapshot is version 0.2.11 at commit `a6f6777586c860457a2a1e6cce6d0bd3fd33d1cc` (Xie, 2025). This is a conditional tool-execution panel; it is not part of the 6,104 clinical ALL units and is not the deferred independent confirmation experiment.

Table 11 summarizes the historical panel.

Table 11: Historical explicit-model validation. Each configuration contains 288 questions from 24 virtual parameter/initial-state profiles. Counts report correct outputs within the supplied model contract.

Config.	Historical arm	Binary /288	Numerical /288	Full output /288
M4	Ordinary reanswer	118	0	0
M4	Full execution	286	270	270
M4	Patch without intervention execution	234	67	67
M4	Typed patch and execution	288	288	288
M5	Ordinary reanswer	187	0	0
M5	Full execution	265	222	222
M5	Patch without intervention execution	234	67	67
M5	Typed patch and execution	288	288	288

The typed patch results demonstrate agreement within the supplied model and question contract. They do not establish new-patient validity, cross-simulator transfer, or unrestricted causal reasoning in clinical text. The present manuscript reuses this historical panel without new language-model calls or numerical integration.

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